CPE 381 Classwork 05: Cubic Spline

Student Name (as in Canvas): _____

A Number: _____

Points: 10 Duration: 10 minutes Due: September 09, 2026 (In-Class)

Consider signal samples as $(t_1, x_1), (t_2, x_2), \cdots, (t_{n+1}, x_{n+1})$.

1. For $n+1$ samples in a signal, if we were to interpolate using cubic spline, how many spline interpolant we would need? **(3 Points)**

Answer

2. What's free boundary conditions in cubic spline formulation? **(3 Points)**

Answer

3. What's the advantage of cubic spline interpolation over linear interpolation? **(4 Points)**

Answer